

Thorben Foerst

DevOps & Cloud Enthusiast

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Summary

IT Management student (B.Sc.) at the Technical University of Cologne, specialising in cloud and DevOps topics. AWS Certified Solutions Architect – Associate, with practical experience in setting up Kubernetes clusters using Terraform and in working with GitOps workflows via ArgoCD.

Skills

Cloud (AWS)

VPC, IAM, EC2, S3, RDS, EKS, Secrets Manager and many more (SAA-C03 certified)

Containers & Orchestration

Kubernetes, k3s, Amazon EKS, Docker, Docker Compose, Helm

IaC & CI/CD

Terraform, GitHub Actions, ArgoCD (GitOps), Ansible, Git

Monitoring & Observability

Prometheus, Grafana, Uptime Kuma

Network & Infrastructure

VLANs & subnetting, DHCP/DNS/firewall configuration, Linux, TrueNAS, reverse proxies (Traefik, Nginx Proxy Manager), load balancing, WireGuard

Programming

Python, HCL, Bash, YAML, JSON, SQL, C#, C++

AI

RAG, LangChain, Chroma, OpenAI API

Certifications

AWS Certified Solutions Architect – Associate

SAA-C03 – Expiry date: 6 May 2029
Credly verification

Languages

German (native language)
English (C1)

Education

B.Sc. in IT Management

TH Köln – Campus Gummersbach

Oct. 2022 – expected Feb. 2027

- Bachelor's thesis: *"GitOps vs. Traditional CI/CD in the Kubernetes Context: A Comparative Evaluation Study Using ArgoCD and GitHub Actions"*

Experience

Software developer

Foerst GmbH

Oct. 2021 – Sep. 2022 & Oct. 2025 – March 2026

- Design and development of interactive training modules for driving simulation software
- Development of an in-house RAG-based chatbot (Streamlit, OpenAI API, ChromaDB) with a knowledge base of 330 pages, used by the development department to assist with the creation of new exercise modules and, according to feedback from the team, saving time in day-to-day development work
- Containerisation and deployment of applications on on-premises servers using Docker and Docker Compose
- Induction and professional support for new staff

Projects

Home Lab

Personal project

June 2025 – ongoing

- Setting up a custom-designed network with its own routing and VLAN segmentation as the basis for the entire infrastructure
- Virtualisation layer on two Proxmox nodes with 6 VMs; running a Kubernetes cluster (k3s) with 2 master and 2 worker nodes to ensure resilience in the event of individual node failure
- GitOps deployment via ArgoCD for 11 services in production (including Nextcloud, Immich and Vaultwarden), Longhorn storage with automated S3 backups
- Infrastructure-as-Code using Terraform, and monitoring with Prometheus and Grafana

Practical project work

TH Köln, module in the final semester, basis for the Bachelor's thesis

Apr. 2026 – Aug. 2026

- Setting up a production-grade Kubernetes infrastructure in the AWS Cloud using Terraform, including the deployment of a managed Kubernetes cluster (Amazon EKS)
- Containerisation and deployment of a test application with a database connection (Amazon RDS) for end-to-end validation of the infrastructure
- Secure management of database access credentials using AWS Secrets Manager, and provisioning via AWS Load Balancer Controller (LBC)